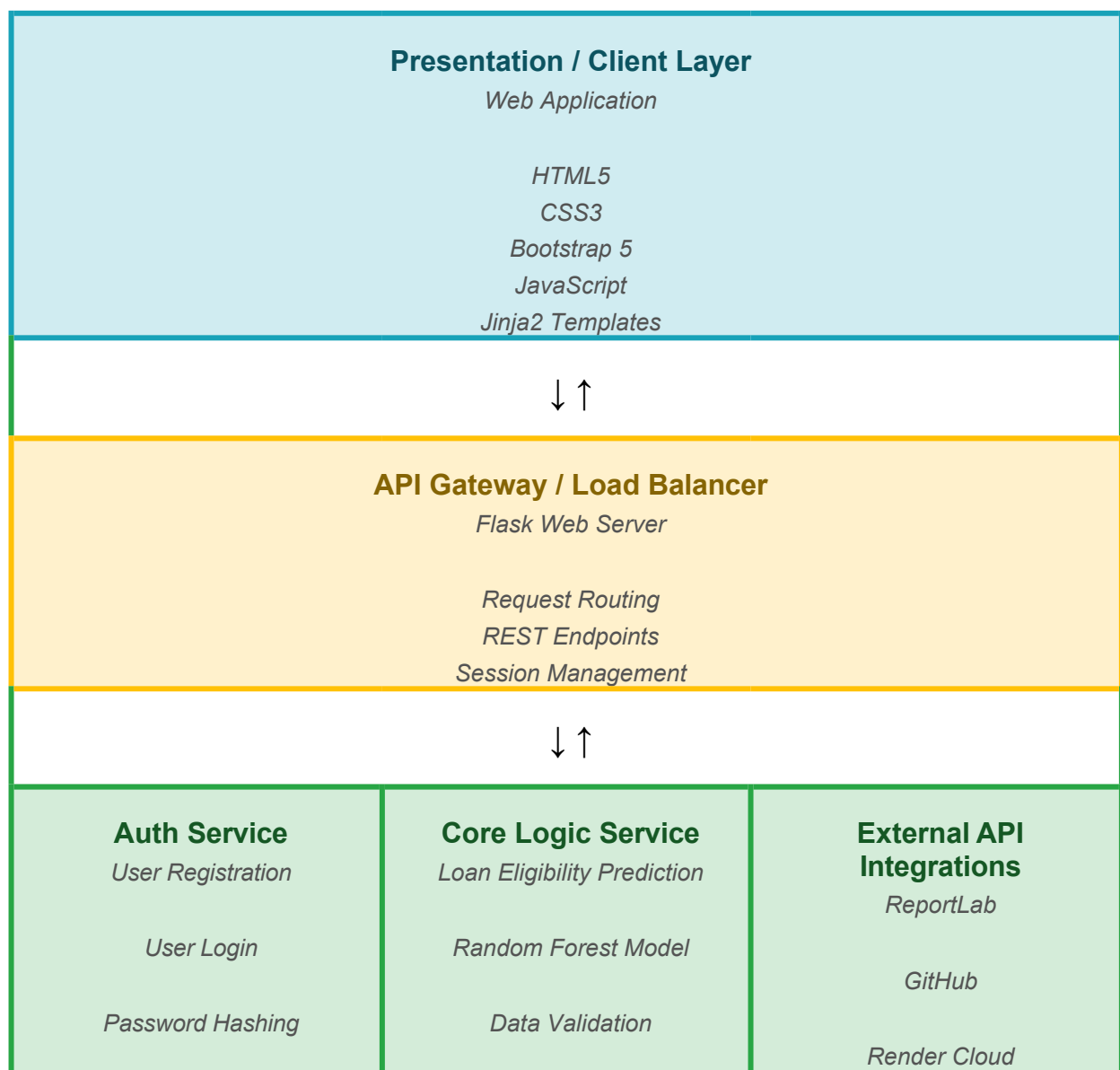


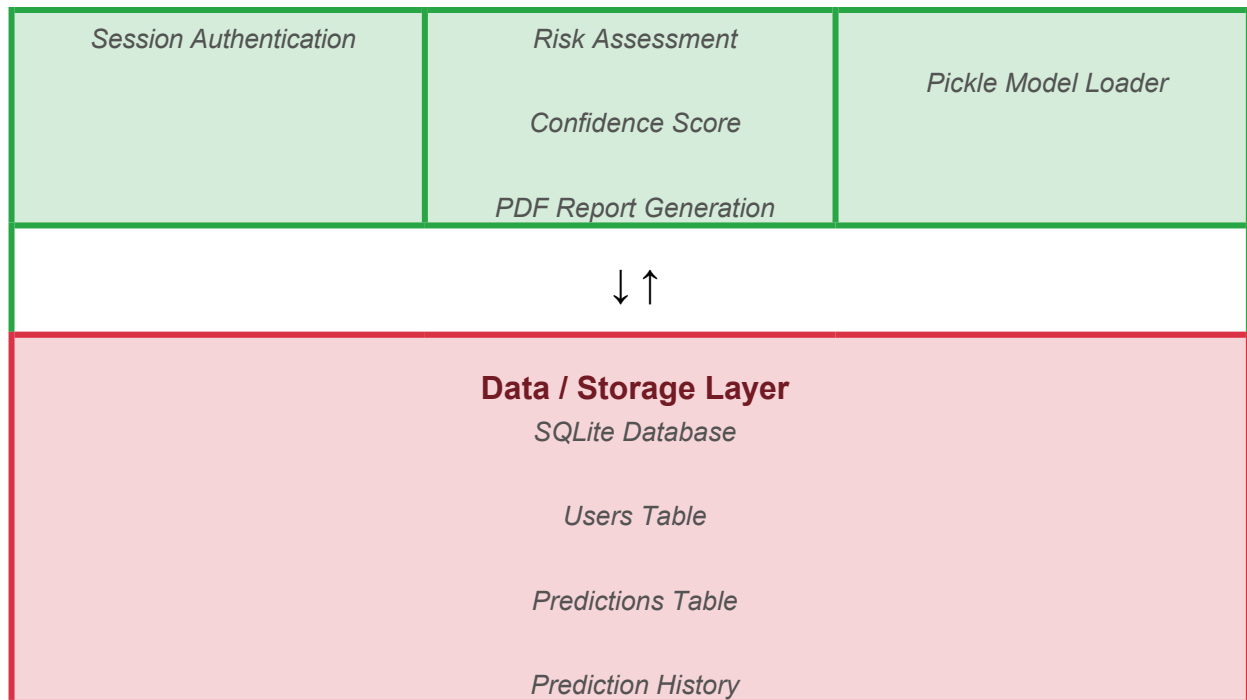
Solution Architecture

Date	29 June 2026
Team ID	
Project Name	Smart Lender: AI-Powered Loan Eligibility Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.





Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a responsive web interface where users can register, log in, submit loan applications, view predictions, dashboard analytics, history, and download PDF reports.	HTML5, CSS3, Bootstrap 5, JavaScript, Jinja2
API Gateway	Receives client requests, routes them to the appropriate services, manages sessions, validates requests, and returns responses.	Flask
Core Logic Service	Performs loan eligibility prediction using the trained Random Forest model, calculates confidence score and risk level, stores predictions, and generates PDF reports.	Python, Scikit-learn, NumPy, Pandas, Pickle, ReportLab
Database	Stores user accounts, applicant details, prediction results, confidence scores, risk levels, and prediction history securely.	SQLite

Architecture Flow (for the diagram)

